

Paying for Accuracy

Kinetic proofreading, cost accounting, and what dissipation adds to a mortal learner's toolbox

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Abstract

A catalyst makes a reaction faster without being consumed, and for that reason it cannot make a reaction more *selective* than the binding energies already allow. A proofreading ladder is consumed, and can. The difference between Michaelis–Menten kinetics and Hopfield–Ninio kinetic proofreading is exactly the difference between a step that is restored and a step that is spent, and the slogan that summarises it is that *kinetic proofreading turns energy into accuracy*.

This note observes that the companion notes already carry that distinction, in a form nobody has used. In the weighted graph-structured lambda theories of MeTTaIL, a catalytic step is a *context rule* — it restores its left-hand side, contributes a multiplicative geometric factor, and incurs no primitive cost — while a proofreading step is a *base rule*, which consumes its redex and must clear the funding judgement of the cost-accounting monad. So the Michaelis–Menten bound is a proposition about rule classes, and the price of accuracy is denominated in the same tokens as everything else the mortal scientist buys.

We recall both mechanisms, map them into the graded **where**-clause framework, and derive an exchange rate: error falls as a *power* of the tokens burned, with an exponent that depends only on single-stage kinetics and not on the depth of the ladder. What is scale-invariant about proofreading is therefore not the mechanism but the price of accuracy, which is why the motif recurs wherever computations compose.

We then ask what this adds to a mortal learner. Three things. It supplies the missing account of *commitment*, the seam between weighing a candidate and acting on one. It shows that *energy substitutes for memory*: a proofreading ladder is a sequential likelihood-ratio test whose accumulator has been replaced by position on the ladder, and the correct candidates it throws away are the register it declined to buy. And it comes with a limitative result: a proofreading step is a move in the fibre, so *energy buys confidence and never scope*. All figures are computed by an accompanying script; three levels of learning — one recognition event, one scientist, one population — are worked, and one negative result is recorded rather than smoothed over.

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1 Introduction

The mortal scientist of the companion notes has a proposal mechanism and an evaluation mechanism. It can move in a lattice of hypotheses, and it can run an assay whose verdict is confirm, refute, or bottom. What it has never had is an account of *commitment*: when a graded verdict is permitted to become a crisp act, how reliable that act is, and what the reliability costs.

Biology solved this problem twice, and the two solutions are worth putting side by side, because their difference is the whole content of this note. The first solution is catalysis, described by Michaelis and Menten in 1913 [8]. The second is kinetic proofreading, described independently by Hopfield [4] and Ninio [10] in the mid-1970s. A catalyst speeds a reaction up without being consumed. A proofreading ladder consumes something at every rung, and in exchange achieves a selectivity that no catalyst can reach.

Kinetic proofreading turns energy into accuracy.

That slogan is the thesis. What this note contributes is that in the framework of the companion notes the slogan is not a metaphor but a statement about rule classes and a ledger, with an exchange rate that can be computed.

1.1 What is claimed

- (1) **The Michaelis–Menten/proofreading distinction is already present in MeTTaIL**, as the distinction between context rules and base rules, and the cost-accounting framework already says that the first is free and the second must be funded (§3). Proposition 3.2 states the resulting bound; §6.1 verifies it numerically.
- (2) **The exchange rate is a power law, not an exponential** (Proposition 4.1). Error falls as $T^{-\eta}$ in tokens per accepted product, and η is a property of one stage (Proposition 4.3).
- (3) **What is scale-invariant is the rate, not the mechanism** (§4.3). Exponent and spend are both additive under the calculus’s two composition operators, so their ratio is what survives composition.
- (4) **Three additions to the learner’s toolbox** (§5): a priced account of commitment; the substitution of energy for memory, with a computed crossover; and the limitative result that energy buys confidence and never scope.
- (5) **One negative result**, recorded in §4.4: token accounting does supply an interior optimum where a rate-only account does not, but the optimal ladder is long, and real proofreading ladders are one or two rungs. The holding charge tested here does not close that gap.

All numbers are produced by `kpr.py`, which uses only the standard library and is seeded. §7 gives the ladder in rholang.

2 Two mechanisms, recalled

2.1 Michaelis–Menten

An enzyme E binds a substrate S , forms a complex, and releases a product:



Writing k_{on} and k_{off} for the binding and unbinding rates and k_{cat} for catalysis, the quasi-steady-state treatment of Briggs and Haldane [3] gives the familiar rate law

$$v = \frac{v_{\text{max}}[S]}{K_M + [S]}, \quad K_M = \frac{k_{\text{off}} + k_{\text{cat}}}{k_{\text{on}}}.$$

Two features matter here, and only two.

First, the enzyme is *restored*. It appears on both sides of the overall reaction; nothing about it is used up. Second, and in consequence, catalysis cannot change where the reaction is going, only how fast it gets there. A catalyst lowers a barrier in both directions equally.

The consequence for discrimination is the one this note is about. Suppose the enzyme faces a right substrate and a wrong one, differing only in how tightly they bind, with dissociation rates

d_r and $d_w = f d_r$ for some $f > 1$. Then at equilibrium the ratio of occupancies, and hence the ratio of products made, is $1/f$. Elaborating the catalytic machinery does not improve this. *The equilibrium discrimination factor f is a ceiling, and no amount of catalysis reaches past it.*

Third feature, worth noting because it returns in §6.2 in a completely different place: the rate law *saturates*. Past $[S] \gg K_M$, more substrate buys nothing.

2.2 Kinetic proofreading

Hopfield [4] and Ninio [10] asked how biological copying achieves error rates far below what binding energies allow — protein synthesis discriminates roughly a hundred times better than the amino-acid binding free energies predict — and gave the same answer. Insert, between recognition and commitment, one or more intermediate states, reached by a step that is *driven and irreversible*, and from which the complex can still dissociate.

The construction requires three things, and all three are load-bearing:

- (a) a **branch point**: from the intermediate the complex may either proceed or fall off;
- (b) **irreversibility**: the forward step must not run backwards, which is what forces the coupling to a source of free energy (in the cell, hydrolysis of a high-energy phosphate bond);
- (c) a **reset**: a complex that falls off returns to the start, not to the previous rung, so that the next attempt is a fresh test.

Because each rung is a fresh test with the same bias, the biases multiply. For a ladder of N tests, the error is f^{-N} in the limit where the forward step is slow compared with dissociation. That limit is not free: making the forward step slow is exactly what makes most *correct* substrates fall off too, and the yield collapses along with the error.

McKeithan [7] recognised the same ladder in T-cell receptor signalling, where a sequence of phosphorylation steps discriminates ligands by how long they stay bound; the mechanism is not a curiosity of the ribosome. The energy–speed–accuracy structure of the trade has been studied extensively since, notably by Murugan, Huse and Leibler [9] and, for sensory adaptation, by Lan and coauthors [6].

2.3 The differences, tabulated

	Michaelis–Menten	Kinetic proofreading
The apparatus is	restored	consumed
Thermodynamically	at or near equilibrium	driven, dissipative
Discrimination	bounded by f	f^N , unbounded in N
What is spent	nothing	one quantum of free energy per rung
What is lost	nothing	yield: correct substrates are discarded too
Saturates in	substrate	<i>energy</i> , once a bypass exists (§6.2)
In MeTTaIL	a context rule	a base rule

The last row is the one this note exists to exploit.

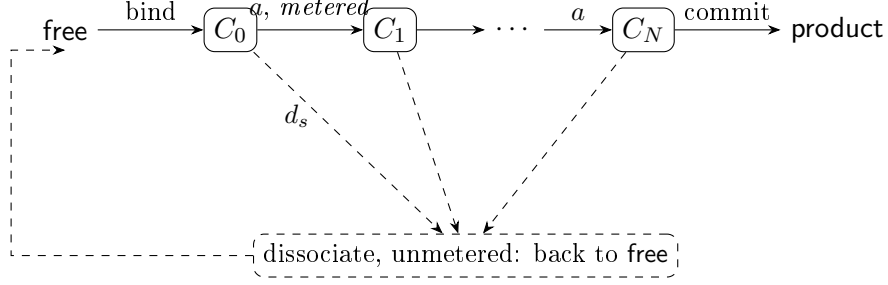


Figure 1: One proofreading ladder. Solid arrows are funded base rules, one token each. Dashed arrows are free: nothing is minted, and the candidate simply ceases to exist as a candidate. The discrimination lives in the race at each rung between a rate that does not depend on the substrate and one that does.

3 Mapping into the graded where-clause framework

3.1 What the framework already supplies

The weighted graph-structured lambda theories of the companion note equip rewrite rules with weight maps whose keys are spatial-behavioural formulae refining the rule's left-hand side and whose values are weights in a commutative semiring. Two rule classes are distinguished there, and the distinction is exactly the one §2 needs.

- A **base rule** $r \vdash (L) \rightsquigarrow (R)$ rewrites its redex. The redex is consumed. In the cost-accounted setting the firing must clear the funding judgement $\mathbf{funds} \Sigma \Delta := \Delta \leq \Sigma$; an unfunded rule has propensity zero.
- A **context rule** $r \mid S \rightsquigarrow T \vdash (\mathcal{C}S) \rightsquigarrow (\mathcal{C}T)$ contributes a multiplicative geometric factor $g(k)$ to the propensity of the rule it wraps, rather than an independent propensity of its own. Its left-hand side is restored. The energetics convention of the implementation states the position plainly: structural rules are addressing rather than dynamics, and incur no primitive cost.

Observation 3.1. A catalyst is a context rule and a pump is a base rule. This is not an analogy that has to be argued for; it is what the two rule forms say.

3.2 The recognition test as a graded clause

A recogniser is a for-comprehension carrying a graded **where**-clause. The clause is a map from candidate matches into the resolution semiring V ; over $V = \mathbb{R}_{\geq 0}$ its values are propensities, and the scheduler resolves the race between enabled redexes in proportion to them.

One rung of a ladder is therefore *two receives on one channel*, whose relative weights are supplied by the clause: an advance arm and a dissociation arm. Writing $\langle K_{\text{adv}} \rangle$ and $\langle K_{\text{off}} \rangle$ for the two context-labelled modalities,

$$\text{Rung}_V(\psi) := \langle K_{\text{adv}} \rangle_V \psi \oplus \langle K_{\text{off}} \rangle_V \perp,$$

and a ladder of N rungs is the N -fold nesting, which is the budget-truncated unfolding of the least fixed point

$$\mu X. ([\beta] \oplus \langle K_{\text{adv}} \rangle_V X),$$

the width-engine schema of the choice-formulae note with the budget stopped at N . The depth of the ladder is a budget on a fixed point, and that is the first place the cost accounting enters: an unbounded μ is an unbounded bill.

3.3 Notation for the rest of the note

Scale rates so that the right substrate dissociates at rate 1 and the wrong one at rate $f > 1$, with a common metered advance rate a . Per rung:

$$p_r = \frac{a}{a+1}, \quad p_w = \frac{a}{a+f}, \quad g = \frac{p_r}{p_w} = \frac{a+f}{a+1} \leq f.$$

g is the discrimination one funded rung buys, p_r the yield it costs.

3.4 The Michaelis–Menten bound, as a proposition about rule classes

Proposition 3.2 (Michaelis–Menten bound). *Let a recogniser’s recognition path be presented entirely by equations and context rules — no metered base rule between binding and commitment. Then its discrimination is at most f , independently of the number of intermediate states.*

Argument. Equations are symmetric and cost-free, and a context rule restores its left-hand side and contributes a multiplicative factor to the propensity of the rule it wraps rather than consuming anything. The internal chain therefore satisfies detailed balance, and the steady-state occupancy ratio of the two species depends only on the binding weights, not on the length of the chain. Multiplying every forward and backward rate on a rung by the same factor leaves the ratio fixed. Only a step that removes the intermediate from the chain without a compensating reverse can make a rung a fresh test. $\square$

Proposition 3.3 (Funded advance multiplies). *If the advance is a base rule that consumes the intermediate, and a dissociated candidate returns to the start rather than to the previous rung, then the discrimination of an N -rung ladder is g^N .*

Proposition 3.2 is verified numerically in §6.1 by solving both presentations on the same chain, and the two columns of Table 1 are the whole argument of this note in numerical form.

Principle 3.4 (Dissipation is the difference). Selectivity beyond the equilibrium factor is bought, not engineered. In this framework the purchase is visible in the syntax: it is the choice of rule class, and the funding judgement is where the bill is presented.

4 Energy is the cost-accounting token

4.1 The ledger

The mortal scientist’s world conserves tokens: $\sigma + \kappa = \Theta$, with σ the purse, κ the spend, Θ fixed, and nothing ever minted. Every metered rewrite debits the purse, and a rule whose debit exceeds the purse does not fire — it is not slowed down, it has propensity zero.

This is the currency in which the price of accuracy is now denominated. There is no need to import a thermodynamics: the ladder is spending the same tokens the forager spends on food and the reproducing scientist spends on offspring, and the three can therefore be compared.

4.2 The exchange rate

Charge one token per advance. Tokens are charged for candidates that are later discarded, and for wrong candidates as well as right ones, since both are entering. Then per candidate entering, the expected number of advances is $p(1 - p^N)/(1 - p)$, and the yield of accepted right candidates is p_r^N , so the tokens per accepted right product are

$$T(N) = \frac{1}{p_r^N} \left[\frac{p_r(1 - p_r^N)}{1 - p_r} + \frac{p_w(1 - p_w^N)}{1 - p_w} \right], \quad \varepsilon(N) = g^{-N}, \quad Y(N) = p_r^N.$$

Proposition 4.1 (The exchange rate is a power law). *As N grows, $\ln T(N) \sim N \ln(1/p_r)$ while $\ln(1/\varepsilon(N)) = N \ln g$. Hence*

$$\varepsilon \sim T^{-\eta}, \quad \eta = \frac{\ln g}{\ln(1/p_r)}.$$

Error falls as a power of the tokens burned per accepted product.

Remark 4.2. This corrects a natural first guess. Counted per *attempt*, accuracy does look exponentially cheap, because the spend per attempt is bounded — most candidates are discarded early and never cost much. But a learner does not eat attempts, it eats accepted products, and per acceptance the yield loss compounds against it. *The whole trade lives in yield*, and any account that prices attempts rather than acceptances will overstate what dissipation buys.

Proposition 4.3 (The rate does not depend on the depth). *η is a function of a and f alone. Ladder depth converts tokens into accuracy at this rate; it does not change the rate.*

4.3 What is scale-invariant is the rate

Kinetic proofreading is usually reported to be scale-invariant by exhibition: the same motif appears in cascade networks, in receptor discrimination, in neural coincidence detection, in courtship, and in the promotion procedures of organisations. Proposition 4.3 suggests a better statement.

The calculus builds terms with exactly two composition operators, prefixing and parallel composition. Both the discrimination exponent and the token spend are additive under both, under the appropriate independence conditions. Their *ratio* is therefore invariant under composition.

Principle 4.4 (The price of accuracy is what recurs). What is scale-invariant about kinetic proofreading is not the mechanism but the exchange rate. The motif appears wherever computations compose because composition is exactly what the rate is indifferent to.

The two independence conditions are different, and the framework can state both. *Serial* independence is freshness: the advance must leave no correlated residue in the new intermediate, which is the reset condition (c) of §2.2 and which the companion notes already discuss as freshness by quotation. *Parallel* independence is graded separation $\varphi_i \mid_0^N \varphi_j$, the factorisation theorem of the composition note. Where either fails, the shortfall is measurable, and §6.3 measures it.

Proposition 4.5 (Only composition multiplies). *A single rung's discrimination is bounded by f however much is spent on it: $g(a) = (a + f)/(a + 1) < f$ for all $a > 0$, with the bound approached only as $a \rightarrow 0$, i.e. as the yield goes to zero.*

This is why ladders exist at all, at every scale. A recogniser that wants better than f has no option but to compose, and composition is the only operation available that multiplies.

4.4 An optimum, and a negative result

Because tokens are charged for discards, and discards grow with depth, there is an interior optimum in the pair (a, N) for any accuracy target. This is already a contribution of token accounting: a rate-only account has no principled stopping point, and a token account does.

The honest report, however, is that the optimum found here is a *long* ladder — five to ten rungs for targets between 10^{-2} and 10^{-4} — whereas real proofreading ladders are one or two rungs. A per-rung holding charge was tested, on the reasoning that an intermediate occupying a rung is structure that must be kept and that keeping structure is metered. It does not close the gap: raising the holding charge raises the cost without shortening the ladder, because the higher yield available at a faster advance dominates. Whatever makes biological ladders short is not this term. §6.1 records the numbers, and this is listed as an open problem rather than resolved.

5 What this adds to the learner’s toolbox

5.1 Commitment, which the toolbox was missing

The learner has a proposal mechanism (revision moves in the relaxation lattice, with an ultrametric geometry and no gradient) and an evaluation mechanism (the assay, with its trichotomy of confirm, refute, and bottom). It has had almost nothing on *commitment*.

That gap is exactly the seam the choice-formulae note names: a graded clause is a quantifier, a map $\text{Cand} \rightarrow V$, whereas a scheduler is a selection function, a map into Cand . The note says that the lift from the first to the second is where the oracle lives, and prices the altitude of the lift, but says nothing about its *reliability*.

A proofreading ladder is that price. It sits precisely between weighing and acting, it is made of the same clause language, and it converts a bounded discrimination into an arbitrary one at a stated rate.

Observation 5.1. The swarm already exhibits the resulting division of labour: dance evaluation is the clause, piping and takeoff are the selection, no bee does both, and quorum sensing is the ladder in between [11]. What was described in the quorum note as a cost argument for sensing at the site rather than at the cluster is a proofreading argument: the irreversible commitment is paid for at interaction price.

So the assay is not replaced. It becomes rung one of something that has a depth dial, and the dial has a price.

5.2 Energy substitutes for memory

This is the part that seems to reach furthest.

Wald’s sequential probability ratio test [12] achieves error ε in about $\ln(1/\varepsilon)/\ln g$ samples by accumulating a log-likelihood ratio. It needs a register, and because it has one, an unlucky early sample is recoverable: the accumulator simply moves back.

A proofreading ladder achieves the same scaling with *no accumulator at all*. The evidence is held in position on the ladder, write-once. Nothing is summed, nothing is stored, and an unlucky sample is fatal — the candidate is gone.

Principle 5.2 (The ratchet and the register). Kinetic proofreading is the sequential likelihood-ratio test with the accumulator replaced by a ratchet. The correct candidates it discards are the register it declined to buy.

This is a statement the framework can make quantitative, because holding structure is already priced: the depth-graded schedule charges for the formulae a learner keeps. Let m be the price per step of holding the accumulator, so a sequential test costs $\ln(1/\varepsilon)/\ln g \cdot (1 + m)$ while the ladder costs $T(N)$. Equating gives a crossover memory price, and §6.2 computes it.

The reading is that **a learner too poor in memory to store its evidence can buy the same discrimination by dissipating instead** — and, in the regime where memory is expensive, should. That is a genuine addition to the toolbox: an instrument that trades one of the learner’s scarce resources for another, in a framework where both are denominated in the same tokens.

5.3 Energy buys confidence, never scope

Now the limitative result, and it is the one to state loudly.

The hypothesis space of the omnibus is a fibration: an ultrametric base of formulae, and a V -valued fibre over each. A revision move changes the formula, and so moves in the base. A proofreading rung changes the value assigned to a candidate, and so moves *only in the fibre*. A fibre move has base-distance zero.

Proposition 5.3 (Confinement is not relieved by dissipation). *A proofreading ladder is a fibre operation. Consequently no amount of token expenditure on proofreading moves a learner out of the ball it started in, and the Confinement proposition applies to a well-funded learner exactly as it applies to a poor one.*

Principle 5.4 (Energy buys confidence, never scope). A well-funded science becomes maximally certain of the wrong ontology, and the certainty is correctly earned. Nothing has gone wrong; the tokens really did buy discrimination. They bought it inside a fibre.

This has the same shape as the topen-boundary result of the quorum note — a shape error is unreachable by refinement — and it is a contribution to the question of why scientists get misled rather than to the ecology proper. It also predicts an asymmetry worth looking for: richer scientific communities should show *sharper* disagreement across paradigm boundaries rather than convergence, because each side is spending its surplus on confidence within its own fibre.

5.4 Two walls, one on each side

Proofreading amplifies a discrimination the learner must already have. At $f = 1$ no amount of energy helps: $g = 1$ for every a , and the ladder returns exactly what it was given. That is the same statement as the forager’s bootstrap trap, where a value of zero is absorbing and a continuously enabled receipt never fires.

So the toolbox now has a clean division of labour, with a wall at each end:

	Exploration	Proofreading
Does what	manufactures a discrimination	amplifies one that exists
Efficiency	cannot be efficient	power law in tokens
Moves in	the base	the fibre
Wall	the bootstrap trap ($f = 1$)	the Michaelis–Menten bound

Neither substitutes for the other, and a learner needs both. What proofreading does *not* add is anything at all about direction or distance in the lattice. Most of learning is search rather than discrimination, and this instrument is entirely on the discrimination side.

6 Three levels of learning

6.1 Level 1: one recognition event

Catalysis against dissipation. Both presentations are solved on the same $(N+1)$ -state chain: the catalytic presentation with reversible internal steps and dissociation only from C_0 , the proofread presentation with irreversible advance and dissociation from every rung. Table 1 is Proposition 3.2 and Proposition 3.3 side by side.

N	catalytic (restored)	proofread (consumed)
0	0.10000900	0.10000900
1	0.10000899	0.01089207
2	0.10000898	0.00118626
3	0.10000897	0.00012920
4	0.10000896	0.00001407

Table 1: Discrimination achieved, $f = 10$, $a = 0.1$. The catalytic column is flat at $1/f$: adding steps to a presentation that restores its intermediates buys nothing whatever. The residual 9×10^{-6} is the finite product drain used to make the steady state well posed.

The exchange rate. Table 2 gives error, yield and tokens per accepted product for $f = 10$, $a = 0.1$, where $g = 9.1818$ and $p_r = 0.0909$. The predicted exponent is $\eta = \ln g / \ln(1/p_r) = 0.924655$; the measured local slope between $N = 7$ and $N = 8$ is 0.924655.

N	error	yield	tokens/acceptance	$\ln(1/\varepsilon)/\ln T$
1	1.089×10^{-1}	0.090909	1.109×10^0	21.4477
2	1.186×10^{-2}	0.008264	1.321×10^1	1.7181
3	1.292×10^{-3}	0.000751	1.463×10^2	1.3341
4	1.407×10^{-4}	0.000068	1.610×10^3	1.2011
5	1.532×10^{-5}	0.000006	1.772×10^4	1.1333
6	1.669×10^{-6}	0.000001	1.949×10^5	1.0922
7	1.818×10^{-7}	—	2.144×10^6	1.0647
8	1.980×10^{-8}	—	2.358×10^7	1.0449

Table 2: The exchange rate. Error falls as a power of tokens burned per accepted product; the last column converges to η from above.

The rate belongs to one rung. Varying a at fixed $f = 10$: $\eta = 0.5811$ at $a = 0.02$, 0.7419 at 0.05, 0.9247 at 0.1, 1.4115 at 0.3, 2.4594 at 1.0, 4.0971 at 3.0. Per factor of accuracy, many weak rungs are cheaper than a few strong ones. Note that this is a statement about a only; N does not appear.

The optimum, and the negative result. Minimising tokens per acceptance over (a, N) for a target error, with a per-rung holding charge c :

c	target	best a	best N	tokens/acceptance	yield
0	10^{-2}	1.336	3	6.52	0.1870
0	10^{-3}	1.803	5	16.21	0.1101
0	10^{-4}	4.437	10	32.82	0.1310
1	10^{-2}	4.437	5	22.61	0.3619
1	10^{-3}	5.989	9	47.93	0.2492
1	10^{-4}	4.437	10	79.91	0.1310
4	10^{-4}	4.437	10	221.19	0.1310

The optimum is interior in both parameters even at $c = 0$, which is the positive finding. The ladder it prefers is long, and the holding charge does not shorten it. Recorded as open problem [O1.](#)

Cross-check. Direct Monte Carlo at $f = 10$, $a = 1$, $N = 3$, four hundred thousand trials per species, seed 20260817: simulated error 0.006005 against closed form 0.006011; yield 0.125305 against 0.125000; tokens per acceptance 7.7931 against 7.7994.

6.2 Level 2: one scientist

The bottom rate is a bypass. A Hopfield ladder is destroyed by a bypass — a path to commitment that skips the rungs. Discrimination then saturates at the bypass fraction, however long the ladder.

A mortal scientist has one, and it is already named in the corpus. The assay that returns bottom because the *observer* is underfunded is indistinguishable from the assay that returns bottom because the *participant* is stuck. A scientist that must act anyway is taking the bypass. Writing b for the fraction of candidates that commit on bottom, the error is

$$\varepsilon_b(N) = b g^{-1} + (1 - b) g^{-N},$$

with floor b/g .

b	floor	N^* (within 10% of floor)	tokens at N^*
0	—	—	—
0.01	1.089×10^{-3}	5	1.772×10^4
0.05	5.446×10^{-3}	4	1.610×10^3
0.20	2.178×10^{-2}	3	1.463×10^2
0.6104	6.648×10^{-2}	2	1.321×10^1

Table 3: Past N^* the tokens keep being spent and the accuracy does not move. The two anchors $b = 0.6104$ at purse 60 and $b = 0$ at purse ≥ 200 are measured elsewhere in the corpus; intermediate values are swept, not measured.

Principle 6.1 (Michaelis–Menten at both ends of the ladder). With a bypass present, accuracy per token falls to zero at finite depth: the exchange rate *saturates in energy* the way the Michaelis–Menten rate law saturates in substrate, and the bottom rate is playing the role of K_M .

Corollary 6.2 (Poverty caps accuracy, with a rate attached). *Since b is a decreasing function of the purse, the maximum discrimination available to a learner is bounded by its wealth, independently of how many tokens it is willing to spend on the ladder. A poor learner cannot buy its way*

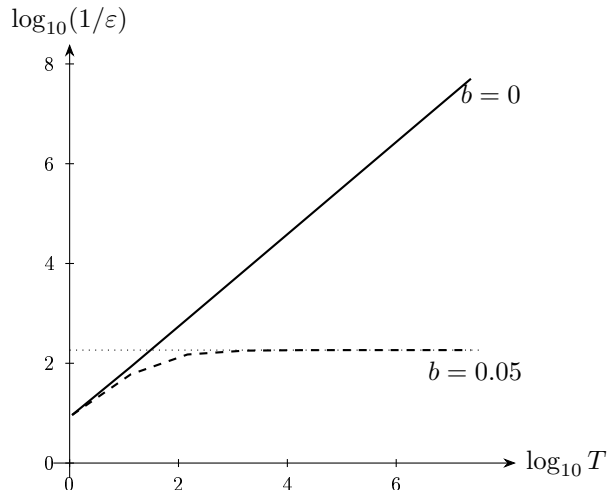


Figure 2: The exchange rate, with and without a bypass. Without one, accuracy is a straight line in tokens on log–log axes, of slope $\eta = 0.9247$. With a bypass at five per cent, the curve saturates: past four rungs the tokens buy nothing. The ceiling is set by b and cannot be raised by spending.

to accuracy even if it spends everything, because what it is short of is the ability to tell a refutation from an underfunded observation.

The ratchet against the register. Giving the sequential test exactly the evidence one rung extracts, so that the comparison is between bookkeeping schemes rather than between tests:

target	ladder N	ladder tokens	sequential samples	crossover price m
10^{-2}	3	1.463×10^2	2.077	69.4
10^{-3}	4	1.610×10^3	3.115	515.9
10^{-4}	5	1.772×10^4	4.154	4263.7
10^{-6}	7	2.144×10^6	6.231	344019.5

The crossover price is where the ratchet becomes the cheaper instrument, and it climbs steeply with the accuracy demanded. The fraction of *correct* candidates discarded at these settings is 0.999249, 0.999932, 0.999994 and 1.000000 respectively — and that column is the honest statement of what the ladder paid.

Observation 6.3. At this advance rate the register is enormously the better buy unless memory is hundreds of times more expensive than a rewrite. That is a substantive prediction, not a disappointment: it says that a learner with any usable memory at all should accumulate rather than ratchet, and that ratcheting is the instrument of learners whose state cannot be held — which is exactly the situation of a molecular recogniser, and exactly the situation of a population whose members cannot share a register.

6.3 Level 3: a population

A population is a converter too. Each reviewer is a rung; rejection is the discard arm; the tokens are the reviewers’ own. The question the framework can now ask sharply is whether the exchange rate survives aggregation.

Reviewers are modelled as weak tests calibrated to the same (p_r, p_w) , with a shared latent component of weight k and a private component of weight $1 - k$: $k = 0$ is disjoint namespaces, $k = 1$ is one reviewer wearing m hats. The effective exponent is $n_{\text{eff}} = \ln(1/\varepsilon)/\ln g$, and n_{eff}/m is the fraction of burned tokens that became discrimination.

k	$m = 1$	$m = 2$	$m = 4$	$m = 8$
0.00	0.9969	0.9949	1.0028	1.0026
0.25	1.0039	0.8962	0.7545	0.5825
0.50	0.9961	0.7948	0.5862	0.4022
0.75	1.0066	0.6958	0.4471	0.2788
1.00	0.9948	0.4992	0.2490	0.1246

Table 4: n_{eff}/m , the efficiency of the population as a converter. Two hundred thousand trials per cell, seed 20260818, reviewers calibrated to $g = 1.5$. At $k = 0$ the reviewers multiply and every token converts. At $k = 1$ the column is $1/m$ to three figures: the other $m - 1$ reviewers bought nothing at all.

Principle 6.4 (Overlap is a tax on the exchange rate). Correlation between stages is not a combinatorial nuisance but a thermodynamic inefficiency, and n_{eff}/m measures it. A population of correlated reviewers is a lossy converter, and the loss is computable from the grading vector.

Observation 6.5 (A replication crisis is an efficiency collapse). The observable signature is specific: adding reviewers continues to cost and stops discriminating. It is not that the reviewers are bad — each row of Table 4 has the same $m = 1$ entry — but that they are not independent stages. The remedy the framework suggests is not more review but disjoint namespaces, and the grading vector says what that means.

Finally, Proposition 4.5 in numbers. Against m disjoint rungs giving 1.089×10^{-1} , 1.186×10^{-2} , 1.292×10^{-3} , 1.407×10^{-4} , 1.669×10^{-6} , 1.980×10^{-8} for $m = 1, 2, 3, 4, 6, 8$, a single rung with the advance slowed m -fold gives 1.089×10^{-1} , 1.045×10^{-1} , 1.030×10^{-1} , 1.022×10^{-1} , 1.015×10^{-1} , 1.011×10^{-1} — creeping towards $1/f = 0.1$ and never past it. Spending more on one rung does not multiply. Only composition multiplies.

7 The ladder in rholang

The listing below is written in the conventions of the companion notes: **new** is sugar, intermediates are manufactured by quotation, and a parameter written **c** is a name while **@d** is data.

```
// -----
// One rung. The two arms are TWO RECEIVES ON ONE CHANNEL, so the
// scheduler resolves a race between them, and the graded where-clause
// supplies the weights. Only the advance arm is metered.
// -----
contract rung(@k, cand, up, waste, budget) = {
  // advance: a funded base rule. The gate {% P %}[s] fires only if
  // the budget stack can cover the debit; unfunded means propensity
  // zero, not a slower rate.
  for (@c <- cand where bindsAt(c, k)) {
    {% up!( quoteFresh(c, k) ) %}[ advance ]
  }
}
```

```

|
| // dissociate: unmetered. The intermediate is dropped, and the
| // candidate must start again from the bottom. Nothing is minted
| // and nothing is restored: this arm is where irreversibility lives.
| for (@c <- cand where bindsAt(c, k)) {
|   waste!(c)
| }
|
| // -----
| // The ladder. Rungs are composed in parallel over a chain of
| // manufactured names, so the ladder is literally the N-fold
| // composition of one rung.
| // -----
| contract ladder(@n, inp, out, waste, budget) = {
|   match n {
|     0 => for (@c <- inp) { out!(c) }
|     - => {
|       // a name fresh in the current scope, by quotation
|       new mid in {
|         rung!(n, *inp, *mid, *waste, *budget)
|         | ladder!(n - 1, *mid, *out, *waste, *budget)
|       }
|     }
|   }
| }
|
| // -----
| // The recogniser. Perception must not be able to act: the arm that
| // READS a candidate and the arm that ADVANCES it are on different
| // namespaces, or looking at a candidate would promote it.
| // -----
| contract recognise(@n, source, verdict, budget) = {
|   new inp, out, waste in {
|     for (@c <= source) { inp!(c) } // peek, not consume
|     | ladder!(n, *inp, *out, *waste, *budget)
|     | for (@c <- out) { verdict!("commit", c) }
|     | for (@c <- waste) { Nil } // the discard is silent
|   }
| }

```

Two points about the listing are the substance rather than the syntax. First, the gate `{% P %}[advance]` on the advance arm and its absence on the dissociation arm is the entire Michaelis–Menten/proofreading distinction, written down. Second, the separation of the reading namespace from the advancing namespace is forced, and for the same reason it was forced in the game note and the quorum note: an observer that can advance a candidate by looking at it is not observing.

8 What is shown, what is not, and what is open

8.1 Shown

- (1) The Michaelis–Menten bound holds as a statement about rule classes (Proposition 3.2), and the two presentations differ numerically by four orders of magnitude at four rungs (Table 1).
- (2) The exchange rate is a power law in tokens per acceptance with exponent $\eta = \ln g / \ln(1/p_r)$, verified to six figures (Proposition 4.1, Table 2).

- (3) The exponent is a property of one rung, so ladder depth converts at a fixed rate rather than changing it (Proposition 4.3).
- (4) Only composition multiplies discrimination (Proposition 4.5, §6.3).
- (5) A bypass caps discrimination, and the mortal scientist’s bottom rate is a bypass; accuracy per token reaches zero at finite depth (Table 3).
- (6) Correlated stages are a lossy converter, with $n_{\text{eff}}/m = 1/m$ at full overlap to three figures (Table 4).
- (7) Token accounting supplies an interior optimum in (a, N) where a rate-only account supplies none.

8.2 Not shown

The claim that a proofreading rung is a fibre move (Proposition 5.3) is argued informally from the fibration reading of the omnibus, which is itself flagged there as assembled rather than established. The ratchet/register comparison prices memory by stipulation; the crossover figures are only as good as that stipulation, and the sensitivity sweep in §6.2 is offered in place of a justification. Nothing here is a claim about biology: the models are the textbook idealisations, and the one place where the idealisation is confronted with the biology — ladder length — it loses.

8.3 Open problems

- O1. **Why are real ladders short?** The optimum computed here is five to ten rungs; biology uses one or two. The per-rung holding charge does not close the gap. Candidates: a fixed cost of installing a rung that is amortised over many recognitions rather than charged per candidate; a constraint from the assembly index of the apparatus; or a yield floor imposed by the ecology rather than by the recogniser.
- O2. **Is the conversion bound the thermodynamic uncertainty relation in a costume?** A bound of the shape $\ln(1/\varepsilon) \leq 1 + \kappa/\text{price}$ should be provable from conservation plus the equation/rewrite split alone. Whether that is a new bound or a restatement of the results of Barato and Seifert [1] and Lan et al. [6] needs a literature pass before any claim is made. The interesting possibility is not the bound but its provenance: a calculus built for concurrency having a second law because it distinguishes equations from rewrites.
- O3. **The quantum rung.** The unitary dilation of the companion note supplies an ancilla, and a discarded candidate is exactly what gets swept into it. Conjecture: the ancilla resource required is the tokens burned. If so, interference cannot buy discrimination for free, which would close the question the weighted-GSLT note left open.
- O4. **Learning the overlap.** n_{eff}/m is measurable by a population about itself. A population that could estimate its own efficiency could reallocate reviewers, which is a graded-clause problem of the kind the composition note poses but does not solve.
- O5. **Does the ratchet ever win in a mortal setting?** The crossover prices computed here are high. Populations whose members cannot share a register are the obvious candidate, and the swarm is the obvious test case.

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